

PROGRAMMING METHODS - LABORATORY 9

Program 01

Use the backtracking algorithm to solve the 0-1 knapsack problem for profit maximization for the example presented below. Show the solution of the problem step by step in the report. Also show the calculations in the form of a text file generated by the program.

Assume $n = 5$, $W = 19$, and the following items are available:

i	p _i [PLN]	w _i	p _i / w _i [PLN]
1	20	2	10
2	30	5	6
3	35	7	5
4	12	3	4
5	3	1	3

General backtracking algorithm for optimization problems

```
void checknode(node v)
{
    node u;

    if (value(v) is better than best)
        best = value(v);

    if (promising(v))
        for (each child u of v)
            checknode(u);
}
```

Laboratory example

Assume $n = 4$, $W = 16$, and the following items are available:

i	p _i [PLN]	w _i	p _i / w _i [PLN]
1	40	2	20
2	30	5	6
3	50	10	5
4	10	5	2

Submission rules

The .cpp file(s) should be named:

Surname_Name_Program_01.cpp

The source files, all other created files, and test text files should be placed in the folder Surname_Name_Laboratory_9.

Submit the .cpp file together with all output text files. The course folder is Programs - lab 9 - Tuesday.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 9 on delta.pk.edu.pl.

Literature

Neapolitan R.: Foundations of Algorithms with Examples in C++, Helion 2004. Additional source from the original sheet: slideshare.net/mandlapure/01-knapsack-using-backtracking.